

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · COMBINED SCIENCE

COMBINED SCIENCE 5006/2

PAPER 2 June 2023

2 hours

PAPER 2 — STRUCTURED QUESTIONS — ANSWER ALL — 2 HOURS

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to June 2023 5006/2.

Additional materials: Answer all questions. The number of marks is given in brackets [].

INSTRUCTIONS: Write your answers in the spaces provided. You may use a calculator.

- 1.** (a) Name two structures found in a plant cell but not in an animal cell. [2]<<
>>(b) State the function of the cell membrane. [2]<<
>>(c) Describe how to prepare a slide of onion epidermis to view under a light microscope. [3]<<
>>(d) Explain why muscle cells contain many mitochondria. [3] **[10]**
- 2.** (a) Name the enzyme in saliva and the substrate it acts on. [2]<<
>>(b) State where bile is made and where it is stored. [2]<<
>>(c) Describe two ways the ileum is adapted for absorption. [3]<<
>>(d) Explain the role of hydrochloric acid in the stomach. [3] **[10]**
- 3.** (a) Name three components of blood. [3]<<
>>(b) State the function of red blood cells. [2]<<
>>(c) Describe how a white blood cell can destroy a bacterium. [3]<<
>>(d) Explain why plasma is needed. [2] **[10]**
- 4.** (a) Name the type of pathogen that causes cholera. [1]<<
>>(b) Describe how cholera is transmitted. [2]<<
>>(c) Explain why oral rehydration is used to treat cholera. [3]<<
>>(d) HIV is a virus. Explain how it leads to AIDS, and state one way transmission can be reduced. [4] **[10]**

5. (a) Describe the arrangement and motion of particles in a solid and in a gas. [4]<<
>>(b) Name the change of state from solid to liquid and from gas to liquid. [2]<<
>>(c) Explain, in terms of particles, why a gas can be compressed but a liquid cannot. [4] **[10]**
6. (a) Name the two substances that must be present for iron to rust. [2]<<
>>(b) Give the chemical name of rust. [2]<<
>>(c) Describe two methods of preventing rusting and explain how each works. [4]<<
>>(d) Why is aluminium slower to corrode in air than iron? [2] **[10]**
7. (a) State the principle of conservation of energy. [2]<<
>>(b) A 2 kg mass is lifted through 10 m. Take $g = 10 \text{ N/kg}$. Calculate the gain in gravitational potential energy. [3]<<
>>(c) Name the energy transfer that occurs in a battery-powered torch when it is switched on. [2]<<
>>(d) Explain how heat is transferred through the metal wall of a kettle by conduction. [3] **[10]**
8. (a) Draw the circuit symbols for a cell, a resistor and an ammeter. [3]<<
>>(b) State Ohm's law. [2]<<
>>(c) A resistor of 4Ω is connected to a 6 V supply. Calculate the current. [3]<<
>>(d) State one difference between a series and a parallel circuit. [2] **[10]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).